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Ultralow and tunable thermal conductivity of parylene C for thermal insulation in advanced packaging

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ABSTRACT

Parylene C thin films have significant applications in advanced packaging of microelectronics. Their thermal properties are critical for thermal management of electronic devices. However, a unified understanding of the tunable structure and the corresponding thermal conductivity is still missing. This study investigated parylene C thin films of varying thicknesses and annealing temperatures (200 ℃ and 320 ℃) grown via thermal chemical vapor deposition. The ultralow thermal conductivity of as-deposited parylene C measured by time-domain thermoreflectance (TDTR) is 0.09–0.11 W/m-K. The thermal conductivity can be tuned by annealing. A significant increase in thermal conductivity is observed in the samples annealed at 320 ℃ (0.18–0.21 W/m-K) which induces melting and recrystallization. The results of XRD, polarized Raman spectroscopy, and GIWAXS show that the enhanced thermal conductivity is attributed to improved crystalline quality and molecular chain reorientation. The measured thermal conductivities of the as-deposited and 200 ℃-annealed films are much lower than the values predicted by Cahill's minimum thermal conductivity model, which can be explained by the diffuson-mediated minimum thermal conductivity model. Parylene C is found to possess the lowest thermal conductivity among dense low-k materials. Our work provides guidance for the structural design of ultralow thermal conductivity polymers and the corresponding thermal design of electronics.

1. Introduction

Parylene C, a semi-crystalline polymer renowned for its exceptional dielectric properties [1,2], mechanical toughness [3], and ability to form ultrathin conformal films [4], has become an important functional coating in microelectronics [5], ranging from complementary metal-oxide-semiconductor (CMOS) devices [6] to micro-electro-mechanical systems (MEMS) [4,7]. Moreover, in three-dimensional integrated circuits (3D-ICs), both thermal insulation and a low dielectric constant are important for electronics packaging [8]. However, when deposited as a thin film, parylene C often exhibits structural and thermal properties that deviate significantly from its bulk counterpart. Despite its widespread applications, the complexity of its nanoscale film structure has left a gap in the comprehensive understanding of thermal transport, particularly regarding how chain structure governs thermal conductivity.

Fundamentally, thermal transport in polymers originates from atomic vibrations propagating through the molecular network, which is bonded by covalent bonds and van der Waals bonds, contributing to the intra-chain and inter-chain heat transport, respectively [9,10]. The thermal resistance of a strong covalent bond is approximately an order of magnitude lower than that of a van der Waals bond [11]. Consequently, the primary bottleneck for thermal transport in polymers lies in the inefficient inter-chain heat transfer due to the weak van der Waals forces. Bulk polymers typically exhibit low thermal conductivity due to disordered structures and the entanglement of molecular chains, but the thermal conductivity of a single polymer chain is several orders of magnitude higher because of less scattering [12] along the chain axis originating from the stiff covalent bonding [13]. Therefore, manipulating molecular chain orientation has emerged as a powerful route to overcome the inter-chain thermal bottleneck, achieving tunable thermal conductivity in both polymer films [10,14] and fibers [15]. Particularly, semi-crystalline polymers comprise amorphous and crystalline parts. In semi-crystalline polymers, this process is bifurcated: in crystalline

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regions, thermal conductivity is dominated by phonons, constrained by phonon-phonon and phonon-boundary scattering; in amorphous regions, vibrational states are categorized as locons, diffusons, and phonon-like propagons [16,17]. While locons are localized and contribute minimally to heat transfer [18], the extended modes (diffusons and propagons) serve as the primary carriers. The bonding strength of the structure partially determines the extent to which each extended vibrational mode contributes to thermal conductivity [19]. Especially in thin films, the mean free path of these propagons is often limited by the size effect, leading to a reduction in cross-plane thermal conductivity [20,21]. A fundamental understanding of these heat transport mechanisms is increasingly critical, as a recent review indicates that accurate theoretical characterization of thermal conductivity of polymers remains largely dependent on their structure [22].

As a representative semi-crystalline polymer, parylene C possesses a complex structure governed by multiple structural factors, including crystallinity, crystallite size, inter-chain spacing, and molecular chain orientation [23,24]. Kachroudi *et al.* [24] and Kim *et al.* [25] indicated that annealing at temperatures below the melting point increased the degree of crystallinity from 45–60% to 75–85% and crystallite size from 4−6 nm to 8−16 nm. The measured thermal conductivities of parylene C with thicknesses of 210 nm, 440 nm, and 760 nm were 0.95, 0.99, and 0.99 W/m-K, respectively, showing very slight thickness dependence [26]. It was also found that, due to surface confinement, parylene C exhibits an amorphous structure at thicknesses below 40 nm, while crystallinity and crystallite size increase as thickness increases [27]. When the crystallite size is less than the phonon mean free path in the crystalline region, boundary scattering becomes dominant and reduces thermal conductivity [20]. Control over crystallinity and chain orientation is typically achieved via process modulation, including post-deposition annealing and deposition temperature tuning during thermal chemical vapor deposition (CVD) process [28]. Systematic research on the relationship between crystalline quality, molecular chain orientation, and thermal conduction mechanisms in parylene C is lacking.

With the miniaturization of integrated circuits, the distance between high-power and temperature-sensitive components decreases, causing unexpected heat crosstalk and degrading device and system performance. For example, advanced 2.5D packaging brings logic and memory closer to facilitate near-memory computing [29]. However, the high heat flux from high-power logic dies poses a critical threat to the adjacent temperature-sensitive memory. To balance the requirements of high-speed communication and thermal reliability, effective thermal insulation strategies must be implemented within the package to hinder heat transport [30]. Due to the ultralow thermal conductivity of parylene C, it is important to study the heat transfer mechanisms and thermal conductivity modulation methods for thermal insulation.

This work prepared parylene C films of varying thicknesses via thermal CVD and investigated their structural and thermal transport properties, as well as the effects of the post-deposition annealing process. X-ray diffraction (XRD), polarized Raman spectroscopy, and grazing-incidence wide-angle X-ray scattering (GIWAXS) were employed to characterize crystalline quality, including crystallinity, inter-chain spacing, and crystallite size, as well as molecular chain orientation. The picosecond acoustic technique and time-domain thermoreflectance (TDTR) technique were used to measure sound velocities and thermal conductivities, respectively. By correlating film thickness, annealing temperature, and phonon transport behavior, this study reveals ultralow and tunable thermal conductivity in parylene C films, which sheds light on the heat transfer mechanisms in ultralow thermal conductivity polymers.

2. Methods

The parylene C thin films, with target thicknesses of 100 nm, 300 nm, and 1 μm, were deposited on 10 mm × 10 mm highly doped silicon substrates using a standardized PDS 2010 thermal CVD system, utilizing 0.16 g, 0.48 g, and 1.6 g of parylene C dimers, respectively. The deposition parameters included a pyrolysis temperature of 690 ℃, a base pressure of 6 mTorr, and a deposition pressure of 15 mTorr. Post-deposition, the films underwent thermal annealing under a nitrogen atmosphere at temperatures of 200 ℃ or 320 ℃ for a total duration of 2.5 h (30 min ramp-up and 2 h steady-state hold). Subsequent thickness measurements using a profilometer revealed a thickness reduction in the films annealed at 320 ℃, which was attributed to the melting of the polymer's amorphous regions.

To characterize the crystallite size, crystallinity, and inter-chain spacing of the parylene C thin films, XRD measurements were conducted using a Bruker D8 Advance system. Samples were analyzed in their as-deposited state and after being subjected to annealing at 200 ℃ and 320 ℃. A Cu Kα radiation source ($\lambda = 1.5418$ Å) was employed, operating at 40 kV and 40 mA. The XRD patterns were recorded over a 2θ range of 10°–30° with a step size of 0.04°.

Polarized Raman spectroscopy (Horiba/LabRam HR Evolution) was used to analyze the molecular chain orientation of parylene C films. A 532 nm solid-state laser was used as the excitation source. The spectra were collected with a resolution of 1 cm^{-1} over the range of 580–1800 cm^{-1}, focusing on the characteristic peak at ~1608 cm^{-1} (assigned to benzene ring C-C stretching vibration, sensitive to chain orientation) and ~695 cm^{-1} (assigned to C-Cl stretching vibration). Measurements were conducted under orthogonal polarization configurations. The film was fixed on a rotatable stage and Raman spectra were recorded every 10° over a polarization angle range of 0°–180°.

To characterize the evolution of molecular chain orientation along the out-of-plane direction, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were conducted with a fixed incidence angle of 0.1°. A Si (220) crystal monochromator was employed to select an X-ray wavelength of 1.547 Å (8.05 keV). The beam was then focused via a Pt-coated mirror, delivering a photon flux exceeding 1×10^{11} photons/s at the sample position. The 2D scattering data were collected using Pilatus 100K and EIGER X1M area detectors with an exposure time of 30 s. A LaB_6 standard sample was used for measurement calibration.

Time-domain thermoreflectance (TDTR) [31,32] is a pump-probe technique, used to measure the cross-plane thermal conductivity of parylene C thin films. A pump laser heats the sample, while a delayed probe laser monitors the temperature change and is collected via thermoreflectance. The collected signal, which varies across samples and embodies their thermal properties, is fitted to a multilayer heat transfer model [33] to infer unknown thermal parameters such as thermal conductivity. The uncertainty of the TDTR system is estimated to be ±8%. A 5 × objective was used, and the pump and probe spot diameters were 22.2 μm and 9.9 μm. All measurements were conducted at a modulation frequency of 5.68 MHz.

3. Results and discussions

3.1. Tunable structure

The thermal CVD and forming process of parylene C films are shown in Fig. 1. In this study, nine samples were prepared and categorized into three groups: as-deposited parylene C with thicknesses of 90, 260, and 970 nm; samples annealed at 200 ℃ with thicknesses of 90, 257, and 960 nm; and those annealed at 320 ℃ with thicknesses of 50, 160, and 680 nm. Notably, the glass transition point is ~90 ℃, and the melting point is ~290 ℃ [28]. Therefore, annealing at 320 ℃ involves melt-recrystallization.

Fig. 2 shows the XRD and polarized Raman results of all samples. All the XRD patterns are smoothed and baseline-subtracted, as the baseline is consistent across all samples. The displayed polarized Raman characterizations correspond to the peak located at ~1608 cm^{-1}. The XRD patterns without subtraction of the baseline and the polarized Raman characterizations regarding the ~695 cm^{-1} peak of parylene C are

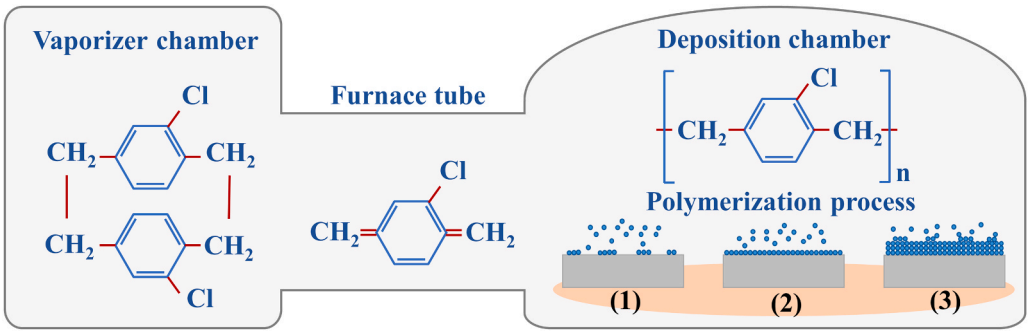


Fig. 1. The deposition process of parylene C thin films via thermal CVD.

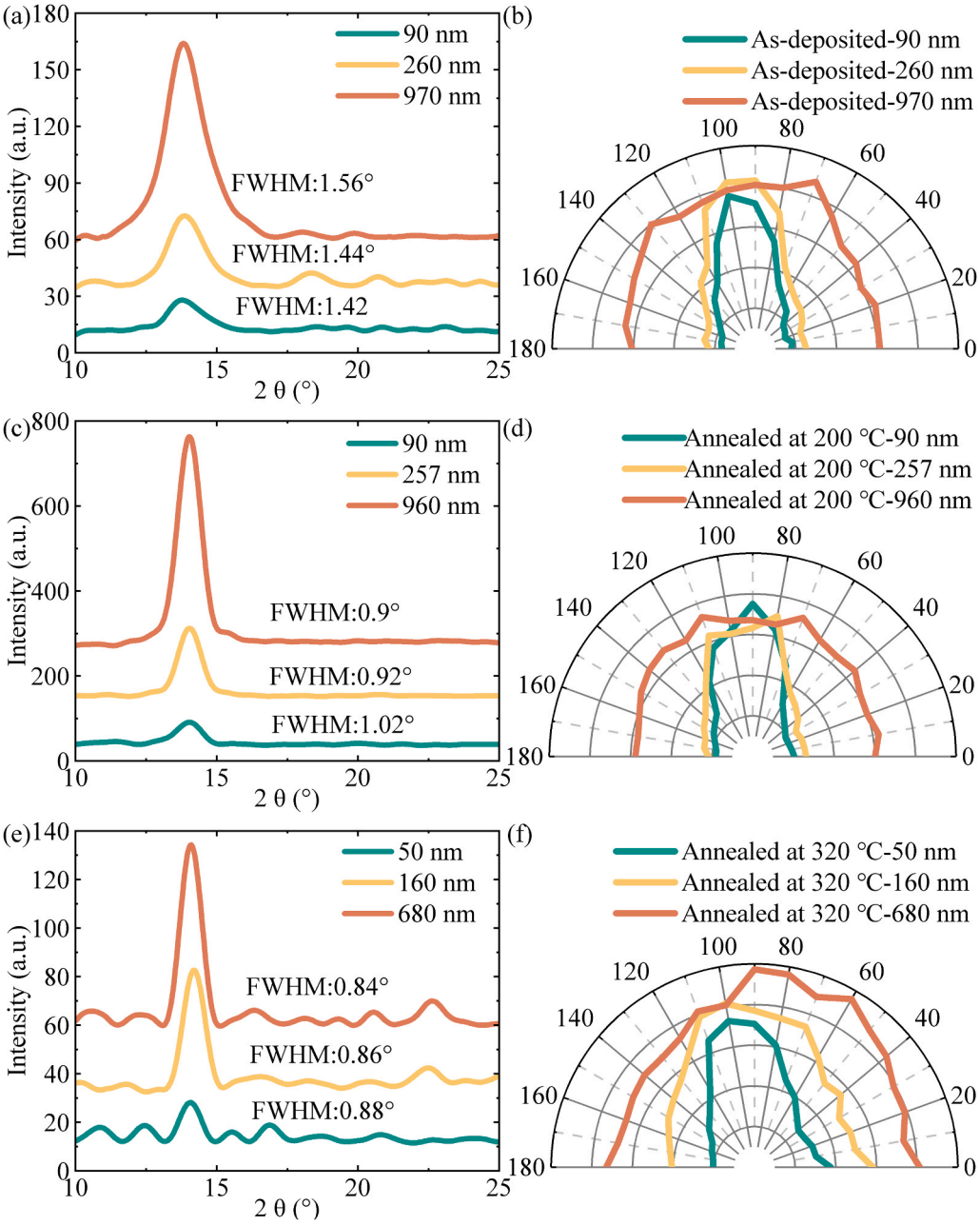


Fig. 2. XRD and polarized Raman characterizations of all samples. (a) XRD patterns and (b) polarized Raman spectra of the three as-deposited parylene C thin films with different thicknesses. (c) XRD patterns and (d) polarized Raman spectra of the three parylene C thin films annealed at 200 °C. (e) XRD patterns and (f) polarized Raman spectra of the three parylene C thin films annealed at 320 °C. FWHM refers to full width at half maximum.

shown in the Supplementary Information. The diffraction peak at ~14° corresponds to the parylene C (020) plane within the monoclinic unit cell [24,27]. Meanwhile, the polarization direction of the laser in polarized Raman spectroscopy varies in a plane parallel to the sample, so parylene C molecular chains align along the *c*-axis and stack in the *b*-axis direction [34]. Namely, molecular chains are parallel to the surface [35].

The XRD patterns in Fig. 2 reveal that annealing results in a narrower full width at half maximum (FWHM), indicating larger crystallite sizes or reduced lattice distortions within the structure. Particularly, annealing at 200 ℃ results in a significant increase in the diffraction peak intensity of the XRD signal, indicating enhanced crystallinity and crystalline quality, which is consistent with the reported work [24]. This effect is more pronounced in thicker Parylene C films. For parylene C annealed at 320 ℃, although the intensity of the (020) peak shows no significant change relative to the as-deposited state, the FWHM is approximately reduced to half that of the as-deposited parylene C, meaning that the average crystallite size becomes roughly doubled based on the Debye–Scherrer formula. During the melt-recrystallization process, molecular chains are freed from the "frozen" state imposed by surface confinement during deposition. The original unstable micro-crystals melt and rearrange to seek the lowest energy configuration. This results in larger crystalline regions that are more stable and have much less distortion. This combination of narrower peaks (larger crystalline regions) with unchanged intensity suggests a mechanism of melt-recrystallization. In addition, both an increase in the annealing temperature (for a given thickness) and an increase in the film thickness (at a given temperature) lead to a systematic shift of the (020) diffraction peak toward higher angles. According to the Bragg relation, this trend indicates a reduction in the inter-chain spacing, which can be observed more clearly in Section S1 of the Supplementary Information. This densification of the structure effectively strengthens the inter-chain van der Waals interactions, which facilitates efficient heat conduction.

The polarized Raman spectra in Fig. 2 reveal a clear thickness-dependent orientation across all sample groups. In thinner films such as those of 90 nm, the pronounced anisotropy in the polar plots suggests that surface confinement forces the molecular chains into a preferential in-plane alignment along a specific direction. However, as the film thickness increases toward the micrometer scale, the curves in polar coordinates become increasingly circular, indicating a transition from anisotropy to isotropy, where the in-plane molecular chain orientation approaches a random distribution.

In addition, the polar coordinate diagrams of 320 ℃-annealed parylene C exhibit a more circular (isotropic) shape compared to the as-deposited and 200 ℃-annealed samples. Annealing at 200 ℃ facilitated minor in-situ rearrangements of the molecular chains. In contrast, annealing at 320 ℃ triggered a phase transition into the molten state, providing the molecular chains with sufficient energy for complete relaxation. Consequently, the melt-recrystallization process effectively eliminates the preferential orientation, yielding a highly isotropic spatial distribution.

It is worth noting that the polarized Raman characterizations can only indicate the in-plane direction variations in the molecular chains. To characterize the evolution of molecular chain orientation along the out-of-plane direction, grazing-incidence wide-angle X-ray scattering (GIWAXS) experiments were performed, as shown in Fig. 3. Both the as-deposited and 200 ℃-annealed parylene C films exhibit highly concentrated and bright diffraction spots along the out-of-plane direction (the q_z axis). Combined with the (020) diffraction peak in the XRD data and the pronounced in-plane anisotropic features in the polarized Raman spectra, these results demonstrate that in these two states, the molecular chains maintain an in-plane distribution along a specific direction. The 200 ℃ annealing merely improves crystallinity but it does not alter the strong anisotropy of the films.

In the GIWAXS pattern of the 320 ℃-annealed parylene C, the previously concentrated out-of-plane (020) diffraction spots and in-plane (110) diffraction spots [36] dissipate and evolve into a continuous Debye–Scherrer ring, aligning with the more circular curve in the polarized Raman spectra. It indicates that the (020) crystal planes are no longer preferentially oriented parallel to the sample surface, with the molecular chains evolving from highly anisotropic to globally isotropic across both the in-plane and out-of-plane directions. It provides direct evidence that a portion of the molecular chains tilt or even become perpendicular, thereby facilitating efficient phonon transport along the cross-plane direction through strong covalent bonds rather than the weak van der Waals bonds.

3.2. Sound velocity measurements

Picosecond acoustic technique is used to measure the sound velocity [37,38], which shares an experimental setup with TDTR shown in Fig. 4 (a). An ultrafast pump pulse (a femtosecond laser) heats the sample surface, generating a strain pulse. The strain wave propagates into the sample until it encounters an interface. A portion of the strain wave continues propagating deep into the sample, while another portion reflects back to the sample surface and changes the sample's optical reflectivity, detected by a delayed probe pulse. Consequently, the intensity of the reflected probe signal is altered [39]. Figs. 4(b-c) show the picosecond acoustic echoes of the three samples. As for the samples annealed at 320 ℃, echo 1 is located at 27 ps, and echo 2 is located at 158 ps. Combined with the thickness of the parylene C film, the longitudinal sound velocity can be determined (160 nm/(158 ps−27 ps)/2). Fig. 4(d) shows the longitudinal sound velocities of five samples. As the annealing temperature increases, the longitudinal sound velocity increases, which facilitates phonon transport.

3.3. The ultralow and tunable thermal conductivities

TDTR was used to measure the thermal conductivity of the parylene C thin films. Before measurements, an 80-nm-thick Al layer was

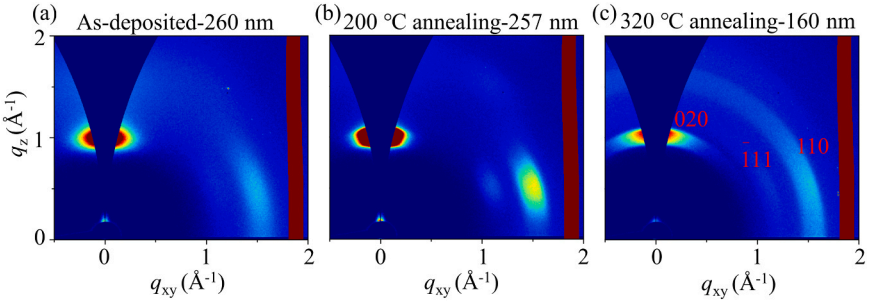


Fig. 3. Grazing-incidence wide-angle X-ray scattering characterizations of (a) the 260-nm-thick as-deposited parylene C thin film, (b) the 257-nm-thick parylene C annealed at 200 ℃, and (c) the 160-nm-thick parylene C annealed at 320 ℃. The vertical red strip on the right side of each image represents the physical gap between the detector modules.

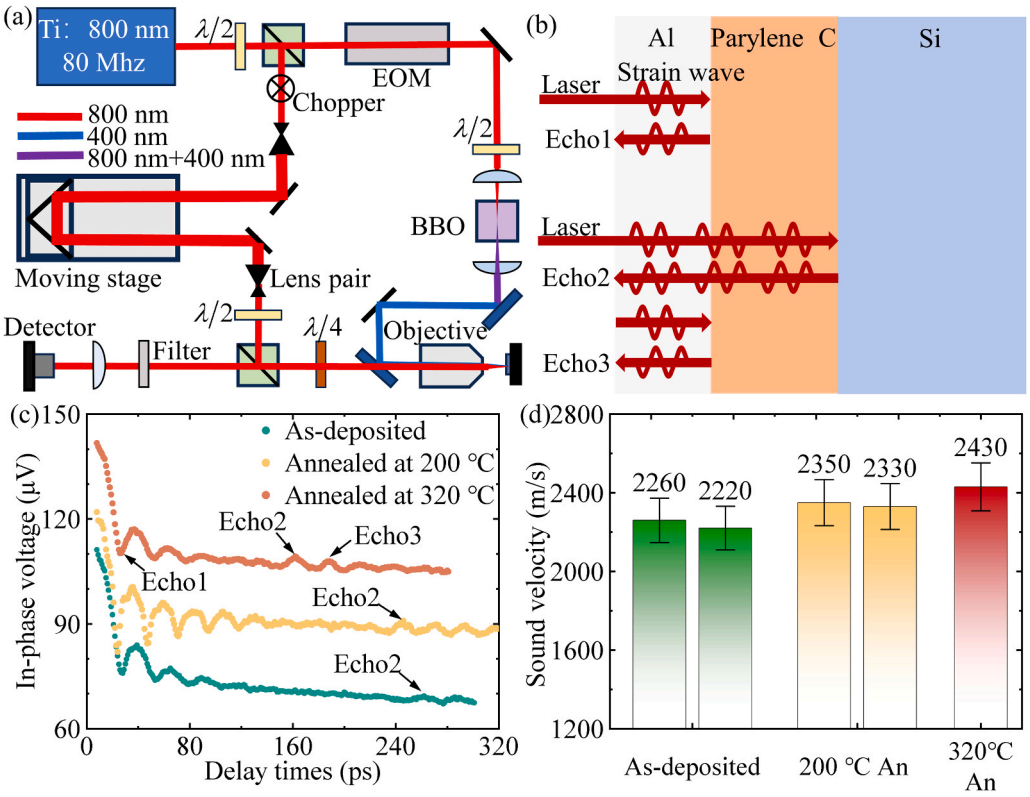


Fig. 4. (a) Schematic diagram of the TDTR experimental setup. The optical setup includes an electro-optic modulator (EOM) for high-frequency beam modulation and a β-barium borate (BBO) crystal for second-harmonic generation. Polarization states are controlled using half-wave ($\lambda/2$) and quarter-wave ($\lambda/4$) plates. The reflected probe beam is collected by a silicon photodetector (connected to a lock-in amplifier) for signal acquisition. A charge-coupled device (CCD) camera is utilized for optical alignment and imaging, which is not included in the schematic diagram. The D-shaped symbol refers to a plano-convex lens used for beam size control. (b) Schematic diagram of the picosecond acoustic mechanism in samples. (c) Picosecond acoustic echoes of the three samples with different thicknesses. The green circles indicate that parylene C has a thickness of 260 nm. The yellow circles correspond to a thickness of 257 nm, while the red ones correspond to a thickness of 160 nm. (d) Longitudinal phonon velocities of the five samples measured by the picosecond acoustic technique. From left to right: as-deposited (90 and 270 nm), 200 °C-annealed (90 and 260 nm), and 320°C-annealed (160 nm) parylene C. 'An' represents the annealed state. Relative error bars of ±5% are employed.

deposited on the samples as a transducer. The sensitivities of TDTR signals to the thermal parameters of the parylene C samples are shown in Fig. 5(a). In the baseline thermal model, the thermal boundary conductance (TBC) values for Al/parylene C and parylene C/Si interfaces were both assigned a typical value of 100 MW/m^2-K. Crucially, even when both TBC values were varied over a wide range from 10 to 500 MW/m^2-K, their calculated sensitivities remained negligible (approaching zero), justifying treating these TBCs as fixed parameters during the fitting process. The high sensitivity to the thermal conductivity of parylene C means it can be measured precisely. The experimental data were fitted with the solution of a multilayer heat transfer model of the sample structure. For the parameters in the data fitting, the thermal conductivity of the Al transducer was measured by the four-point probe method combined with the Wiedemann–Franz law. The thermal conductivity of the highly doped silicon substrate was measured in advance as an input parameter in the data fitting. The heat capacities of the Al transducer [40,41] and Si [41,42] substrate are from the previously reported works. Fig. 5(b) shows an example of the data fitting of TDTR measurements.

Fig. 5(c) presents the measured thermal conductivities of all nine samples. Specifically, the thermal conductivities of the as-deposited parylene C are 0.09 W/m-K, 0.10 W/m-K, and 0.11 W/m-K, while the 200 ℃-annealed parylene C exhibits slightly higher values of 0.10 W/m-K, 0.10 W/m-K, and 0.12 W/m-K, respectively, depending on the corresponding thickness. Under the same annealing conditions, the thermal conductivity increases with film thicknesses, exhibiting an apparent thickness dependence. When the thickness exceeds the phonon mean free path, the intensity of phonon-boundary scattering is reduced. In addition, the transformation of molecular chain orientation emerges as a critical factor. In ultrathin films, a preferential in-plane chain orientation severely hinders phonon transport in the cross-plane direction due to the weak inter-chain van der Waals interactions. In thicker films, the weakening of surface confinement allows for a more randomized molecular chain arrangement, introducing more vertical covalent pathways that facilitate cross-plane heat conduction.

Consistent with the aforementioned XRD and polarized Raman characterizations, annealing at 320 ℃, which corresponds to the melt-recrystallization regime, significantly enhances thermal conductivities (reaching 0.18 W/m-K, 0.21 W/m-K and 0.20 W/m-K depending on different thicknesses). This enhancement is driven by a synergistic mechanism: the improvement in crystal quality—characterized by enlarged crystallite sizes and tighter chain spacing—extends the phonon mean free path and strengthens inter-chain coupling. Simultaneously, the transition to structural isotropy reorients a fraction of the high-thermal-conductivity covalent backbones toward the cross-plane direction. While the former reduces phonon scattering, the latter provides an effective pathway for vertical phonon transport, collectively overcoming the bottleneck of cross-plane inter-chain thermal conduction. In summary, the melt-recrystallization induced by 320 ℃ annealing serves as a critical pathway for optimizing the crystalline structure and enhancing the cross-plane thermal conductivity of parylene C.

Consequently, a significant enhancement in cross-plane thermal conductivity is only realized at 320 ℃, where the melt-recrystallization process mitigates this bottleneck by reorienting covalent pathways toward the cross-plane direction.

Fig. 5(d) shows the temperature-dependent thermal conductivity of

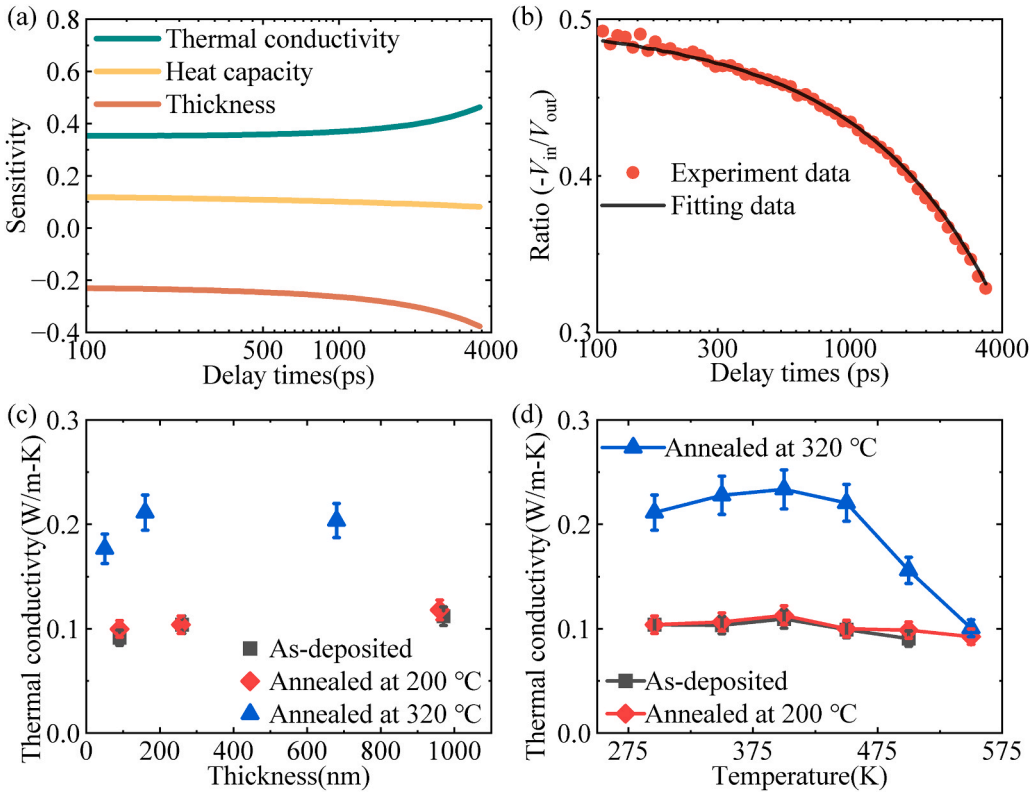


Fig. 5. (a) The sensitivities of TDTR signals to the thermal conductivity, the heat capacity, and the thickness of the parylene C annealed at 200 ℃ with a 90nm thickness. (b) The TDTR data fitting curve: the red circles represent experimental data, while the black line represents the fitting data. (c) The thickness-dependent thermal conductivity of the parylene C thin films under different annealing conditions. (d) The measured thermal conductivity of the parylene C films as a function of temperature. Black squares, red diamonds, and blue triangles represent the 260-nm-thick as-deposited parylene C thin film, the 257-nm-thick parylene C annealed at 200 ℃, and the 160-nm-thick parylene C annealed at 320 ℃, respectively.

the 260-nm-thick as-deposited parylene C, the 257-nm-thick parylene C annealed at 200 ℃, and the 160-nm-thick parylene C annealed at 320 ℃, respectively. Samples were heated using the HCP421V hot plate of Instec company from 296 K to 550 K at 50 K intervals and measured by TDTR simultaneously. The thermal conductivities of the as-deposited parylene C and the annealed parylene C at 200 ℃ are close and remain generally unchanged, while the thermal conductivity of the annealed parylene C at 320 ℃ rises first, and then decreases, which is similar to the polymer of 50% crystalline structure displayed in the reported work [43]. The decrease in thermal conductivity is caused by the intensification of Umklapp phonon scattering within the crystalline domains as the temperature approaches 550 K. Another reason is that as the temperature gradually approaches the melting point, the structure becomes disordered, thereby affecting heat conduction. The heat capacity of parylene C at elevated temperatures used in the TDTR data fitting was measured by differential scanning calorimetry (DSC), which is shown in Supplementary Information.

3.4. Heat transfer mechanisms

To figure out the heat transfer mechanism, the "minimum thermal conductivity" (also called the amorphous limit of thermal conductivity) of parylene C was calculated using Cahill's minimum thermal conductivity model, assuming scattering occurs at a distance equal to the interatomic spacing [44,45]:

$$\Lambda_{\min} = \left(\frac{\pi}{6}\right)^{1/3} k_B n^{2/3} \sum_{i=1}^{3} \nu_i \left(\frac{T}{\Theta_i}\right)^2 \int_0^{\Theta_i/T} \frac{x^3 \exp(x)}{(\exp(x)-1)^2} dx \quad (1)$$

where Θ_i is the Debye temperature; κ_B is the Boltzmann constant; n is the number of atoms per unit volume; ν_i is the sound velocity in the different polarization directions, namely one longitudinal and two transverse.

The Debye temperature is calculated as follows:

$$\Theta_D = \frac{\hbar}{\kappa_B}\left(6\pi^2 \frac{N}{V}\right)^{\frac{1}{3}} \nu = \frac{\hbar}{\kappa_B}\left(6\pi^2 \frac{n_{\text{eff}} \rho N A}{M}\right)^{\frac{1}{3}} \nu \quad (2)$$

$$\frac{3}{\nu^3} = \frac{1}{\nu_L^3} + \frac{2}{\nu_T^3} \quad (3)$$

$$\nu_L = \left[\frac{E(1-\sigma)}{(1+\sigma)(1-2\sigma)\rho}\right]^{1/2} = \left[\frac{2(1-\sigma)}{(1-2\sigma)}\right]^{1/2} \nu_T \quad (4)$$

where $\hbar$ is the reduced Planck constant; κ_B is the Boltzmann constant; N/V is the number of atoms per unit volume; NA is the Avogadro constant; M is the molar mass of parylene C calculated to be 138.59 g/mol using the repeating unit chemical formula C_8H_7Cl; n_{eff} is the effective atomic number of the repeating unit; ρ is the density with a value of 1289 kg/m^3 [26,41]; σ is Poisson's ratio with a value of 0.4 [46]; ν is the average sound speed; ν_L is the longitudinal sound velocity measured by the picosecond acoustic technique and displayed in Fig. 4(d), and ν_T is the transverse sound velocity.

Because hydrogen (H) atoms are not thermally excited at room temperature [47], they are not included in n_{eff}. Simultaneously, C and Cl atoms all lose 0.5 vibrational modes for each bond, and the benzene ring corresponds to two equivalent atoms [48,49]. The n_{eff} can be calculated to be 25/6, which is directly related to heat conduction [50]. Using Eqs. (2–4), the transverse phonon velocity ν_T can be calculated to be 992 m/s when the longitudinal phonon velocity is taken as 2430 m/s, and the Debye temperature of parylene C is calculated to be 96 K, meaning that the atomic vibrations of parylene C are all excited at room temperature.

In our experiments where $T \gg \Theta$(96 K), all vibrational modes are fully

excited. Mathematically, as the upper integration limit Θ_i/T approaches zero, the integral in Eq. (1) can be simplified into a constant. Therefore, Eq. (1) can be simplified as [17,51]:

$$\Lambda_{\min} = \left(\frac{\pi}{48}\right)^{1/3} k_B n^{2/3} \left(\nu_L + 2\nu_T\right) \tag{5}$$

The calculated minimum thermal conductivity using Eq. (5) shown in Fig. 6 is ~0.19 W/m-K. This value is higher than the measured thermal conductivities of the as-deposited and 200 ℃-annealed samples, but remains lower than that of the parylene C annealed at 320 ℃. The enhanced thermal conductivity of the parylene C annealed at 320 ℃ is attributed to its high crystalline quality and the efficient cross-plane covalent pathways, whereas the minimum thermal conductivity corresponds to the amorphous limit. When the temperature exceeds 500 K, the thermal conductivity starts to reduce, gradually approaching the amorphous limit. This is partly because, as approaching the melting point, the material becomes increasingly disordered. Another contributing factor is the enhanced phonon-phonon scattering.

The values calculated by the minimum thermal conductivity model exceed the measured values for the as-deposited parylene C and annealed parylene C at 200 ℃. Agne *et al.* proposed a diffuson-mediated thermal conductivity model which calculates the diffusons' contributions to the thermal conductivity of amorphous materials [52]. The calculation formula is as follows:

$$\Lambda_{\min} = 0.76 k_B n^{2/3} \frac{1}{3} \left(\nu_L + 2\nu_T\right) \tag{6}$$

The calculated thermal conductivities of the as-deposited parylene C and the 200 ℃-annealed parylene C are both ~0.12 W/m-K, consistent with the measured values, shown in Fig. 6. The agreement between the values calculated by the model and the measured thermal conductivities shows that the thermal conduction in the above parylene C samples with smaller thicknesses (such as 270 nm) is dominated by diffusons.

Fig. 6 compares the measured and calculated thermal conductivities of the three parylene C samples. For the as-deposited and 200 ℃-annealed parylene C, the measured values are consistent with k_{diff}, but fall significantly below $k_{\text{diff-full}}$. This confirms that hydrogen atoms

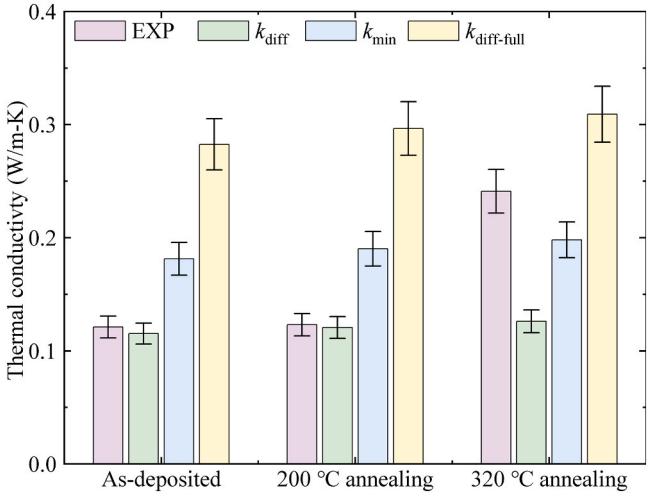


Fig. 6. Comparison of the measured and calculated thermal conductivities of the 260-nm-thick as-deposited parylene C thin film, the 257-nm-thick parylene C annealed at 200 ℃, and the 160-nm-thick parylene C annealed at 320 ℃. EXP is the measured thermal conductivity. $k_{\min}$ and k_{diff} are the calculated thermal conductivities by the minimum thermal conductivity model and the diffuson-mediated thermal conductivity model using the effective atomic number, respectively, while $k_{\text{diff-full}}$ is the calculated thermal conductivity by the diffuson-mediated thermal conductivity model using the full atomic number. The same ±8% relative error bars are employed as for the thermal conductivity measurement by TDTR.

are not thermally excited and do not contribute to the thermal conductivity. It is evident that k_{diff} is 37% lower than $k_{\text{diff-full}}$.

3.5. Thermal insulation

As illustrated in Fig. 7, the thermal conductivity does not strictly correlate positively with the dielectric constant. Notably, among all dense low-k materials, parylene C exhibits the lowest thermal conductivity, nearly an order of magnitude lower than that of traditional inorganic materials such as SiO_2 and fluorinated silicate glasses (FSG), while maintaining a relatively low dielectric constant. This enables the parylene C film to serve not only as a dielectric layer in microelectronic devices [2,53], but also as a thermal insulation layer in the field of integrated circuits, reducing thermal crosstalk between high-power components (e.g., GPU) and temperature-sensitive components (e.g., memory) [30]. Furthermore, parylene C has favorable mechanical strength and structural stability, which is particularly crucial for microelectronic devices and MEMS that require thermal insulation while withstanding mechanical stress [7]. Although porous materials may have lower thermal conductivities and dielectric constants, the poor thermal and mechanical stability poses significant integration challenges in back-end-of-line (BEOL) processes [54,55]. Therefore, parylene C thin films possess distinctive advantages in the fields of advanced packaging, MEMS, and sensors as functional coatings, due to the low dielectric constant, ultralow thermal conductivity, and favorable mechanical strength.

4. Conclusions

This work discussed the relationship between the structure, regulated by thickness and annealing temperature, and the cross-plane thermal conductivity of parylene C thin films at room temperature and under high-temperature conditions. The negligible variations in thermal conductivity of the 200 ℃-annealed parylene C confirm that inter-chain van der Waals forces remain the primary bottleneck of heat conduction along the cross-plane direction. The enhancement in thermal conductivity observed in the 320 ℃-annealed parylene C results from a synergistic structural optimization: melt-recrystallization simultaneously increases crystalline quality and changes the molecular chain orientation, thereby reducing phonon scattering and establishing efficient cross-plane covalent pathways for thermal conduction. The calculated values from the diffuson-mediated thermal conductivity model are

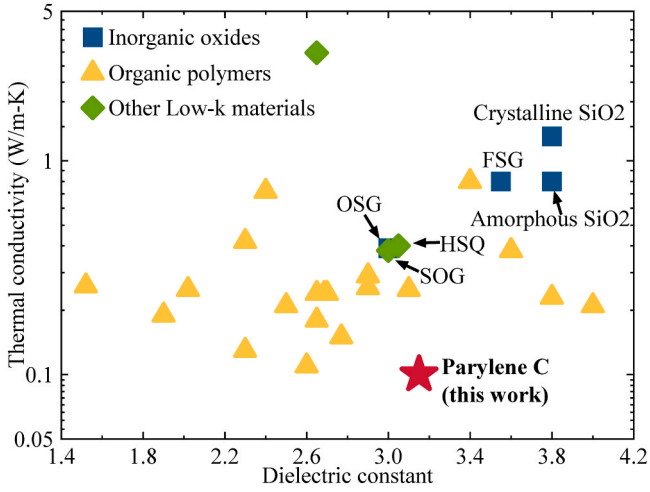


Fig. 7. Comparison of the thermal conductivity of parylene C with that of other dense low-dielectric constant materials, including inorganic oxides, organic polymers, and other low-k materials (organic-inorganic hybrid materials and composites). Detailed information can be found in the Supplementary Information.

consistent with the measured thermal conductivities, indicating that the thermal conductivity is dominated by diffusons for the as-deposited and 200 ℃-annealed parylene C films with smaller thicknesses. Meanwhile, parylene C possesses the lowest thermal conductivity among dense low-k materials. These results offer a comprehensive understanding of heat transport mechanisms and thermal conductivity modulation methods of linear polymers, supporting thermal insulation in advanced packaging.

CRediT authorship contribution statement

Yicheng Wei: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Xingqiang Zhang:** Software, Data curation. **Han Xu:** Writing – review & editing, Writing – original draft, Visualization, Resources, Formal analysis, Data curation. **Zhe Cheng:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Wei Wang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijheatmasstransfer.2026.129326.

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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